

Binomial Distribution



Conditions for a binomial distribution

- 1 State the four conditions required for a random variable to follow a binomial distribution.
 - 2 A survey asks 50 people whether they support a new policy, and X is the number who say yes, assuming each person's response is independent with the same probability of support. Explain why X follows a binomial distribution.
 - 3 A researcher rolls a die 15 times and counts the number of times a 3 appears. Explain why this count follows a binomial distribution.
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Calculating binomial probabilities

- 4 A fair coin is tossed 8 times. Find the probability of getting exactly 5 heads, using $P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}$.
 - 5 A factory produces items with a defect rate of 10%. In a sample of 6 items, find the probability that exactly 2 are defective.
 - 6 A fair coin is tossed 6 times. Find the probability of getting exactly 4 heads.
 - 7 A factory produces items with a defect rate of 20%. In a sample of 5 items, find the probability that exactly 1 is defective.
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Cumulative binomial probabilities

- 8 A biased coin has $P(\text{heads}) = 0.3$. It is tossed 5 times. Find $P(X \leq 1)$, where X is the number of heads.
 - 9 Using the same distribution $X \sim B(5, 0.3)$, find $P(X \geq 2)$.
 - 10 A biased coin has $P(\text{heads}) = 0.2$. It is tossed 4 times. Find $P(X \leq 1)$, where X is the number of heads.
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Mean and variance of a binomial distribution

- 11 A random variable $X \sim B(40, 0.25)$. Find the mean and variance, using $E(X) = np$ and $\text{Var}(X) = np(1 - p)$.

- 12 A random variable $X \sim B(60, 0.2)$. Find the mean and variance.
- 13 A random variable $X \sim B(80, 0.1)$. Find the mean and variance.
- 14 A random variable $X \sim B(50, 0.6)$. Find the mean and variance.
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Finding an unknown parameter

- 15 A random variable $X \sim B(n, 0.4)$ has mean $E(X) = 20$. Find n .
- 16 A random variable $X \sim B(25, p)$ has mean $E(X) = 10$. Find p .
- 17 A random variable $X \sim B(n, 0.25)$ has variance $\text{Var}(X) = 9$. Find n .
- 18 A random variable $X \sim B(n, 0.5)$ has variance $\text{Var}(X) = 12.5$. Find n .
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Using the binomial distribution to model a real scenario

- 19 A multiple-choice quiz has 10 questions, each with 4 options, only one of which is correct. A student guesses randomly on every question. Find the probability the student gets at least 8 questions correct.
- 20 A basketball player has a 70% free-throw success rate. Find the probability that the player makes exactly 3 out of 5 free throws.
- 21 An archer hits the target with probability 0.8 on each shot, independently. Find the probability that the archer hits the target exactly 4 times out of 5 shots.
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Distinguishing binomial situations from non-binomial situations

- 22 Explain why drawing 5 cards from a standard deck WITHOUT replacement and counting the number of aces drawn is NOT a binomial distribution, while drawing 5 cards WITH replacement would be.
- 23 Explain why the number of sixes rolled in 10 rolls of a fair die IS a binomial distribution, while the number of rolls needed to get the first six is NOT.
- 24 Explain why the number of heads obtained by flipping a fair coin exactly 20 times IS a binomial distribution, while the number of flips needed to get 3 heads is NOT.
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A well-designed word problem: quality control sampling

- 25** This question has two parts. A machine produces bolts, of which 5% are defective. A quality inspector randomly samples 20 bolts each hour. (a) Find the expected number of defective bolts in a sample of 20. (b) The inspector rejects the whole batch if 3 or more defective bolts are found in the sample. Find the probability that the batch is rejected, given $P(X = 0) \approx 0.3585$, $P(X = 1) \approx 0.3774$, and $P(X = 2) \approx 0.1887$ for $X \sim B(20, 0.05)$.